Preprint. August 12, 2026.

Rediscovering Relativity Is Not a Single Task

What It Takes for AI to Discover a Physical Theory

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Can an AI system rediscover general relativity? This question is increasingly discussed in the AI community, but it is often framed too narrowly. Some accounts implicitly assume that rediscovery means reproducing Einstein's field equations; others assume that a single batch of experimental data suffices to construct a theory; still others reduce scientific discovery to one mode of inference. We argue that rediscovering a physical theory is not a single task, but a layered sequence of distinct capabilities. Using the history of relativity as a case study, we reconstruct it not as a linear path from Newton to Einstein but as a branching theory space populated by Lorentzian, Poincaréan, scalar, vector, and geometric theories. We show that theory construction and experiment design were mutually generative rather than sequential, and that at several points the available evidence could not adjudicate among live candidates. On this basis we propose a decomposition of rediscovery into distinct capabilities, and three staged tests for AI systems in theoretical physics.

1. Introduction

Can AI rediscover general relativity? This question has recently become a benchmark for evaluating the scientific capabilities of AI systems. One influential proposal is Hassabis's "Einstein test": train a system whose knowledge is restricted to what was known in 1911 and ask whether it can independently arrive at general relativity, just as Einstein did in 1915. Only then, the argument goes, could we claim that the system possesses genuine general intelligence.[1] One such attempt trains models from scratch on corpora containing only pre-1900 texts, aggressively removing any traces of later concepts (Hla, 2026).

The motivation behind these efforts is sound: they try to hide the answer and reduce data contamination. But they still leave the central question underspecified. What counts as "rediscovering" relativity, and how should success or failure be evaluated along the way? This is the gap this paper addresses. In our view, rediscovering relativity is not about recovering a single answer; it is about reconstructing a historical theory space.

So far, the notion of "rediscovery" itself is ambiguous. Does it mean reproducing Einstein's field equation? Does it mean reconstructing the entire logical path from special relativity to general relativity? Or is reproducing only one crucial step sufficient? Different answers correspond to fundamentally different capabilities, yet current discussions often arrive at yes-or-no conclusions without first distinguishing among them.

More importantly, the current discussion often makes the history of relativity look too static, as if all relevant facts had already been collected and the task were to infer the right theory from a fixed archive. The actual history was not shaped in this way. Maxwell's equations predicted electromagnetic waves and incorporated light into electromagnetism, which in turn motivated the search for the so-called "aether wind". The null result of the Michelson–Morley experiment led to Lorentz's contraction hypothesis. The experiments carried out between 1902 and 1904 were no longer testing for the aether wind, but the new predictions generated by the contraction hypothesis itself. The history of relativity was an interaction between theory and experiment: theories suggested what to test next, and experiments forced theories to change. From Maxwell to Lorentz, from Poincaré to Einstein, and later to the competing theories of relativistic gravity, the development of relativity was a continuous process of mutual interaction rather than a one-shot inference.

Instead of discussing rediscovery as one big question, we turn it into three more concrete discovery questions, each closer to the physics and to the historical details:

- Can AI construct and revise theories as new experimental evidence arrives, as Lorentz did?
- Can AI elevate a collection of empirical regularities into a unifying principle, as Poincaré did?
- Can AI construct the entire space of candidate relativistic theories of gravity, derive their predictions, and

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[1] Demis Hassabis, remarks in "Fireside Chat: Sir Demis Hassabis and Prof Govindan Rangarajan," Indian Institute of Science, 17 February 2026, moderated by Varun Mayya. Transcript available at the IISc EECS event page.

identify the experiments that could distinguish among them?

These are not three versions of the same capability. Together, they ask whether a system can move through a theory space, revise that space as experiments arrive, and identify what must be tested next. The boundary is equally important: theory can tell us what would be informative, but not what observation will show.

Our position is that rediscovering relativity requires reconstructing a historical theory space, repeated interactions between theory and experiment, and movement across different modes of reasoning. It cannot be reduced to induction, deduction, or a single act of abduction, nor can it be completed entirely on paper.

Section 2 reconstructs this theory space. Section 3 reviews three common ways of framing the problem. Section 4 decomposes rediscovery into distinct capabilities. Section 5 proposes three staged evaluations. Section 6 explains why the validation of physical theories must ultimately come from outside the theories themselves.

2. The Historical Theory Space of Relativity

2.1. 19th-Century Situations

Newtonian mechanics and universal gravitation had won enormous authority in 19th-century astronomy. The standard example was Neptune: irregularities in the motion of Uranus were used to infer the position of an unseen planet, which was then observed. Mercury's anomalous perihelion precession was a problem, but not yet a reason to abandon Newtonian gravity. The residual was about $43''$ per century, out of a total precession of roughly $5557''$.

The first responses were patches inside the Newtonian picture. Le Verrier proposed an intra-Mercurial planet, Vulcan, in 1859, and astronomers searched for it for decades without confirmation. Other fixes included solar oblateness and Hall's 1894 modification of the inverse-square exponent. These proposals had weaknesses, but Mercury alone did not force Einstein's answer or open up new physics. As long as old models still had plausible adjustments, a patched Newtonian theory remained the natural first move.

Electrodynamics created a different situation. Maxwell's equations unified electricity, magnetism, and light by treating light as an electromagnetic wave. But the equations were not covariant under Galilean transformations, which made the aether look like the natural rest frame for electromagnetic waves. The Michelson–Morley experiment was designed to detect the aether wind, the motion of the Earth through this frame. Its result was null. By the end of the 19th century, relativity was not born from one clean anomaly. It came out of two live situations: a small gravitational residual inside a very successful Newtonian programme, and an electrodynamic theory whose own structure made the aether experimentally suspect.

2.2. Special Relativity: Lorentz, Poincaré, Einstein

Lorentz's theory lived inside the aether framework. His 1895 theory used local time and length contraction to explain why the Michelson–Morley experiment did not detect aether drift. Later Rayleigh, Brace, and Morley–Miller tested consequences of the FitzGerald–Lorentz contraction idea; Trouton–Noble tested whether a charged capacitor moving through the aether should feel a torque. Lorentz had to update his theory with a more elaborate electron model and additional stabilising assumptions to match these new experiments, but he did not abandon the aether. In this sense, Lorentz was still patching a physical picture.

Compared with Lorentz's increasingly complex theory, Poincaré started from a simpler first principle (he called it "the principle of relativity"). In 1904, he treated the impossibility of detecting aether drift as a general postulate. He corrected Lorentz's formulation, showed that the Lorentz transformations form a group, and made the symmetry requirement explicit. Physically, Poincaré still accepted the aether framework and treated the time and length measured in other frames as artificial quantities. Mathematically, this brought his theory very close to Einstein's.

Unlike the other two, Einstein removed the privilege of the aether and the absolute frame. His 1905 work started from the relativity principle and the constancy of the speed of light, without referring to any hidden rest frame. The time t' of a moving inertial frame was not a calculational fiction, as Poincaré suggested, but the real measurable time of that frame. Today this is the physical interpretation we usually identify with special relativity. But in 1905 there was no experiment that could separate Lorentz, Poincaré, and Einstein by their observable predictions. For the experiments at issue, Lorentz's, Poincaré's, and Einstein's theories gave the same observable results, even though they rested on different physical pictures.

2.3. Theory and Experiment as Mutually Generative

We clarify the interaction between theory and experiment. Between 1887 and 1902, there was only one aether-drift experiment - the Michelson–Morley experiment. Lorentz's 1895 theory of length and local time was good enough to explain the Michelson–Morley result. However, the next experiments were different. Rayleigh and Brace tested whether FitzGerald–Lorentz contraction should produce double refraction in moving matter. Morley and Miller tested whether the effect depended on the material construction of the interferometer. Trouton–Noble tested whether a charged capacitor moving through the aether should feel a torque. These were not just more searches for aether wind. They were new questions created by the theory itself.

All of them gave null results. This is a typical example of theory and experiment moving together: Lorentz's theory explained one experiment, then inspired new experiments, and those experiments pushed him from contraction alone toward the full Lorentz transformation and a more complicated electron theory. After that, no new decisive experiment arrived. Poincaré and Einstein then rebuilt the same experimental situation in cleaner ways, each simpler than Lorentz's physical picture, but neither reformulation was selected by a new observation.

2.4. Relativistic Gravity Before General Relativity

Special relativity is naturally described in Minkowski spacetime, but Newtonian gravity does not fit there. Poisson's equation is not Lorentz covariant. How should gravity be brought into Minkowski spacetime? Between 1905 and 1915 this was an open problem worked on by Poincaré, Nordström, Abraham and Mie, not by Einstein alone. The standard move was to keep spacetime flat and add a gravitational field on it, exactly as one adds an electromagnetic field. The field has to be Lorentz covariant, it has to reduce to Newtonian gravity in the static weak-field limit, and its source has to be the energy-momentum tensor $T_{\mu\nu}$.

Poincaré took the vector in 1905, modelling gravity on electromagnetism, where the source is a four-current j^μ and the field a four-potential A^μ. It was the only template available, since the energy-momentum tensor did not yet exist as a tool. It gave gravitation a propagation speed c and what Poincaré called gravitational waves, but little else to test with observation: a perihelion advance for Mercury well short of the observed $43''$, and no other prediction.

Nordström kept Newton's potential as a scalar and kept it as a field on Minkowski spacetime. Newton's equation contains derivatives in space only, which is why its gravity acts everywhere at once. Nordström added the missing time derivative, which turns it into a wave equation and makes the potential travel at c, and he replaced mass density as the source by the energy of matter, since in relativity the two are the same thing. His theory was self-consistent and attractive, and it explicitly requires $m_i = m_g$, so the equivalence principle was not Einstein's private concern, and the gravitational redshift comes out right for that reason. However, he calculated a perihelion precession of $-7''$ per century for Mercury, with the sign reversed, and no bending of light passing through the gravitational field of matter.

Einstein started on the same flat background. From 1907 to 1912 he worked with a scalar as well, using a position-dependent speed of light $c(x)$, and the $0.83''$ deflection he published in 1911 came from it. He gave it up when it could not be carried beyond static fields or made to hold the full equivalence principle, and turned to geometry.

No principle available before 1915 selects the second-rank tensor. The argument that a massless spin-2 field is the only consistent option requires the representation theory of the Poincaré group, which Wigner published in 1939. Einstein reached the tensor by a route on which the choice never arises. Once gravity is taken to be the geometry of spacetime rather than a field placed on it, the field has to be the metric, and the two indices of $g_{\mu\nu}$ in $ds^2 = g_{\mu\nu}dx^\mu dx^\nu$, together with their symmetry, follow from the quadratic form itself. The rank was not selected. It was handed over by the geometry.

What separates the candidates is narrow. Redshift does not: Nordström's value is correct. Mercury is not clean either, since it requires non-linear corrections and had been measured since 1859. The discriminating quantity is light deflection, 0 against $1.75''$, and it fixed the rank of the field. It did not fix the physical picture behind it. A massless spin-2 field on a flat background, coupled to $T_{\mu\nu}$ and to its own energy, sums to the Einstein equations with $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, a result obtained by Gupta in 1954, Kraichnan in 1955, Feynman in 1962, and Deser in 1970. That reading and curved spacetime remain empirically indistinguishable. Fixed constraints, one open choice, and one observable that settles it: this is what makes the episode statable as a task, and we return to it in Section 5.3.

3. Common Framings of Rediscovery

3.1. The Whig Framing

- The framing: rediscovery means arriving at Einstein's theory. Operationally, an evaluation scores a system on whether it outputs the Einstein field equations, or the two postulates of special relativity.

- Where it appears: informal benchmark proposals; popular framings of the question; historical narratives that follow Einstein alone and omit Nordström, Poincaré, Abraham, Mie.

- What it presupposes: that at each historical moment there was a determinate correct answer available.

- Cite specific instances.

3.2. The One-Shot Framing

- The pipeline: experimental data $\rightarrow$ AI $\rightarrow$ physical theory. Keep the existing diagram.

- Its concrete form: "give a system all of physics up to 1905 and have it produce general relativity."

- Where it appears: public statements by AI laboratory leadership; benchmark designs that supply a complete corpus in a single prompt.
- What it presupposes: that the evidence base is fixed and prior to theorising.

3.3. The Single-Mechanism Framing

- The framing: discovery reduces to one mode of inference.
- Its three current versions: compression/induction, Schmidhuber; deduction from given axioms, the formal-mathematics line, AlphaProof; a single abductive jump requiring embodied simulation, Zahavy 2026.
- What it presupposes: that one mechanism accounts for the whole episode.

4. Rediscovery Is Not One Task

- Problem recognition. Which facts require explanation? $m_i = m_g$ had zero residual and was known for two centuries. Mercury's $43''$ invited parameter adjustment. The magnet–conductor asymmetry produced no discrepancy at all.
- Sequential theory construction. Build from incomplete evidence; revise as observations arrive. Include the 1913 fact here: the Entwurf failed the general covariance Einstein himself required and gave $18''$ for Mercury, while Nordström's theory was complete and self-consistent. Any search pruned on intermediate performance discards Einstein's branch.
- Principle formation. From accumulated compensations to a single principle; the group structure as the mathematical expression of "no further patch is possible."
- Hypothesis-space construction. Generate multiple viable theories; avoid premature collapse onto one.
- Internal screening. Lorentz covariance, conservation, stability, Newtonian limit, and the one screening that works on paper: vector gravity gives repulsion between like charges.
- Recognising empirical equivalence. Distinguish mathematical difference from observational difference. Note that adjudication at the gravity stage settled the field type, not the ontology.
- Experimental discrimination. The decisive quantity is light deflection, $1.75''$ vs 0. Redshift does not discriminate: Nordström's is correct. Mercury requires non-linear corrections and is not a clean criterion.
- World-coupled adjudication.

5. Three Staged Tests

- These are conceptual tests, not benchmarks ready to run. The answers are present in any modern training corpus, so success is uninformative; only failure is informative, and only counterfactual variants, altered source symmetry or altered spacetime dimension, restore any diagnostic value.

5.1. The Lorentz Test

- Input, staged: first Maxwell plus aether plus Michelson–Morley alone; then Rayleigh/Brace, Trouton–Noble, Morley–Miller.
- Tasks: construct an explanation; derive its new predictions; on receiving the second batch, decide whether to patch, abandon, or reformulate.
- What would count as success: deriving from one's own hypothesis the predictions that the second batch will test; recognising that those predictions have been refuted; tracking the growth in auxiliary assumptions.

5.2. The Poincaré Test

- Input: the full set of null results and Maxwell's equations. Withheld: the relativity principle, the Lorentz group, Einstein's postulates.
- Tasks: find a principle that explains all results at once; derive the admissible transformations; establish closure.
- What would count as success: the group property, and an explicit statement of why a principle differs from a sequence of compensations.

5.3. The Gravity Test

- Input: Minkowski spacetime; Lorentz covariance; $\partial^\mu T_{\mu\nu} = 0$; recovery of Newtonian gravity in the static weak field. Withheld: equivalence principle, Riemannian geometry, spin-2 gravity, Einstein, general relativity.
- Tasks: enumerate candidate field types with justification; write couplings and field equations; compute Newtonian limit, perihelion precession, light deflection, redshift; separate paper-eliminable from observation-requiring candidates; name the decisive observation.
- What would count as success: the enumeration is complete, spin 0, 1, 2; vector is eliminated without computation; the decisive observable is identified as light deflection with the reason why redshift and perihelion do not serve.
- Table of the three candidates and their predictions.

Table 1. Candidate relativistic gravity theories and discriminating predictions.

CANDIDATE FIELD TYPE	MERCURY PRECESSION	LIGHT DEFLECTION	GRAVITATIONAL REDSHIFT	OUTCOME
SPIN 0, SCALAR (NORDSTRÖM)	$-7''$ PER CENTURY, SIGN REVERSED	0, SPACETIME CONFORMALLY FLAT	CORRECT	SURVIVES ON PAPER, ELIMINATED BY OBSERVATION
SPIN 1, VECTOR (POINCARÉ)	ADVANCE WELL SHORT OF $43''$	NONE PREDICTED	NONE PREDICTED	ELIMINATED WITHOUT COMPUTATION, LIKE SOURCES REPEL
SPIN 2, TENSOR (EINSTEIN)	$+43''$ PER CENTURY	$1.75''$	CORRECT	SURVIVES

6. Why Physics Must Remain Coupled to the World

- Formal proof certifies mathematical validity internally; AI for mathematics can in principle close its loop.

- Physics cannot. Simulation does not supply independent evidence: it executes the theory under test. Computing $1.75''$ from general relativity and then citing that number as confirmation of general relativity is circular. The 1919 measurement supplied an input from outside the theory.

- The loop: theory $\rightarrow$ prediction $\rightarrow$ experiment $\rightarrow$ observation $\rightarrow$ revision.

- Execution may be automated, including self-driving laboratories and robotic observatories. The claim is not about who operates the instrument. The claim is that the validating information must originate outside the theory being tested.

7. Discussion and Conclusion

- Restate: the question is not whether a system reaches Einstein's answer.

- The decisive test is whether it can reconstruct the space of possibilities, recognise what remains undecidable on paper, and determine which question must next be put to nature.

- Where the division of labour falls: enumeration and paper-elimination on one side, adjudication on the other.

- Limitations: a single case; the enumeration in Section 5.3 uses representation-theoretic tools unavailable before 1939; the tests are conceptual.

References

Hla, M. Machina mirabilis. `https://michaelhla.com/blog/machina-mirabilis.html`, March 2026. Online research note.

A. Appendix A

- A.1 The three candidate gravity theories: couplings, field equations, and derivations of the four observables.

- A.2 Timeline of experiments and theories, 1859–1919.